Advancing AI Image Forensics: Machine Learning Approaches for Detecting Real and AI-Generated Images Using Multi-Source Unstructured Image Datasets

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Abstract*—The rapid advancement and accessibility of generative artificial intelligence (AI) tools such as MidJourney, Stable Diffusion, DALL·E, BigGAN, Glide, VADM, and WuKong have significantly increased the production of hyper-realistic synthetic images. While these technologies foster creativity and innovation, they also present serious challenges to digital forensics, media authenticity, and cybersecurity, as distinguishing AI-generated images from real ones has become increasingly complex for both humans and traditional forensic techniques. This study presents a robust machine learning framework for detecting AI-generated images across multiple generation platforms using multi-source unstructured datasets. Leveraging a fine-tuned Swin Transformer Tiny (Swin-T) architecture, the proposed system identifies subtle visual cues and generative artifacts that differentiate synthetic content from authentic imagery. The methodology includes standardized preprocessing—such as image normalization, resizing, and corruption detection—and applies temperature scaling and threshold optimization to enhance confidence calibration and classification accuracy. Experimental results demonstrate that the model achieves strong generalization across diverse AI platforms, maintaining high recall while preserving interpretability through Explainable AI (XAI) visualization techniques like Grad-CAM. This research contributes to digital forensics by establishing a reliable, interpretable, and scalable detection framework that strengthens misinformation mitigation and promotes trust in visual media authenticity.

Keywords*—*AI-generated images, digital forensics, image classification, machine learning, convolutional neural networks, ResNet, unstructured datasets, synthetic image detection, deep learning, multi-source datasets, Grad-CAM, Explainable AI, Swin Transformer Tiny

I. Introduction

A. Overview

The rise of generative AI has made it increasingly difficult to distinguish real images from AI-generated ones, creating risks of misinformation, identity theft, fraud, and digital manipulation. Traditional forensic methods that rely on pixel artifacts or compression traces are becoming less effective as generative models advance. This raises the central research problem: can we build a reliable system to detect whether an image is AI-generated or real? Using datasets such as GenImage, machine learning offers a promising approach by training models to identify subtle patterns that differentiate synthetic from authentic content, with the broader goal of enhancing digital forensics, ensuring online trust, and mitigating the societal risks posed by deceptive media.

B. Significance of the Study

The relevance and importance of this research lie in the fact that AI image generation tools such as Midjourney, Stable Diffusion, DALL·E, and Canva AI are now widely accessible, making it increasingly difficult for humans to detect subtle fakes with the naked eye. This creates a pressing need for reliable detection systems to ensure trust and authenticity across critical domains like media, journalism, education, and social media. From a machine learning perspective, the problem is highly significant because it challenges models to go beyond simple classification and instead uncover the "visual fingerprints" of AI-generated content—patterns often invisible to humans. By building a machine learning model capable of distinguishing real from synthetic images, this research not only contributes to digital forensics and online safety but also advances the broader field of adversarial machine learning, robustness, and generalization in detecting artificially generated media.

C. Research Questions and Objectives

The study aims to achieve the following objectives:

1. Develop and evaluate machine learning models (e.g., CNN, ResNet, Swin Transformer) to accurately classify images as real or AI-generated.
2. Assess model performance across multiple AI image generation platforms including ADM, BigGAN, Glide, MidJourney, SDv5, VADM, and WuKong, identifying challenges posed by different generative approaches.
3. Investigate model explainability using Explainable AI (XAI) methods to determine which visual features and artifacts most strongly influence classification decisions.

The research questions guiding this study are:

- How effectively can machine learning models differentiate real images from AI-generated images across multiple platforms?

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- Which AI generation platforms produce images that are most challenging to detect, and why?
- What image features and patterns contribute most to successful classification, and how can they be interpreted to enhance model transparency and explainability?

These questions are **specific, measurable, achievable, relevant, and time-bound (SMART)**, as the study utilized a well-defined dataset, established machine learning algorithms, and clearly measurable evaluation metrics such as accuracy, precision, recall, and F1-score.

II. Background and Literature Review

Forensic Analysis of Image Metadata to Distinguish AI-Generated Images

In this research, a metadata-based method studying the ability to detect AI-generated images by leveraging the Kaggle publicly available DeepDetect-2025 dataset that contains a balanced representation of pictures generated by AI and captured by humans. Through the extraction and inspection of the Exchangeable Image File Format (EXIF) metadata, we find specific patterns of metadata that are unique to images generated by AI systems in comparison with real ones; a mismatch in the camera make/ model fields, compression signature, and lack of geolocation tags are the primary patterns to be considered. Based on the results, fast lightweight and metadata-only forensic approaches to AI image verification are possible, with potential application to automated moderation systems and workflows of an investigation.

Image Classification And Explainable Identification Of AI – Generated Synthetic Images

Recent advances in synthetic image generation, particularly through artificial intelligence, have led to the creation of images so realistic that they are virtually indistinguishable from real photographs. This presents significant challenges for data authenticity and reliability, especially in areas such as journalism, social media, and scientific research, where the integrity of images is critical. This study proposes an approach to effectively distinguish between real and AI-generated images using a deep learning model based on ResNet50. The classification task is framed as a binary problem, where images are categorized as either "real" or "AI-generated." While synthetic images can replicate complex visual details such as lighting, reflections, and textures, subtle visual imperfections often differentiate them from genuine photographs. The study investigates these differences, focusing on minor artifacts and inconsistencies that are typically present in AI-generated content, such as background distortions, lighting anomalies, and unnatural textures. These artifacts are not always perceptible to the human eye, but can be reliably detected by machine learning models.

Classification of Real Vs AI Generated Images Using Deep Learning

This paper presents a Classification of Real vs AI images using Deep Learning With the rise of powerful AI-generated images, spotting the difference between real and synthetic pictures has become more important than ever. Deep learning, particularly convolutional neural networks (CNNs) and transformer-based models, is helping us tackle this challenge. These models analyze tiny details in images—such as textures, patterns, and hidden artifacts—to detect whether an image was created by AI tools like GANs or diffusion models. This technology has practical uses in many areas, from preventing misinformation and verifying images in journalism to strengthening cybersecurity against deep fakes. By training on a diverse mix of real and AI-generated images, these models learn to classify them with high accuracy.

Detection of AI-Generated Images: A Mixed Methods Study on Age-Related Differences

This study uses a mixed methods approach to examine the vulnerability of individuals across differing characteristics. However, there were no differences of age cohort on the rate of hits, misses, or correct rejections, nor on image discernment reaction time while controlling for analytical thinking. The increased false alarm rate among older adults may reflect AI sensitivity toward images, potentially indicating that AI sensitivity acts as a protective factor in the AI interactions of older adult's AI interactions.

AI-Generated Video Detection via Perceptual Straightening

The rapid advancement of generative AI enables highly realistic synthetic videos, posing significant challenges for content authentication and raising urgent concerns about misuse. Existing detection methods often struggle with generalization and capturing subtle temporal inconsistencies. We propose ReStraV(Representation Straightening Video), a novel approach to distinguish natural from AI-generated videos. Inspired by the "perceptual straightening" hypothesis which suggests real-world video trajectories become more straight in the neural representation domain, we analyze deviations from this expected geometric property.

III. Methodology

A. Research Design

This research follows a quantitative, experimental design using supervised machine learning. The study will involve model training, validation, and evaluation to classify images as real or AI-generated. The experimental design ensures reproducibility and facilitates comparison across multiple AI platforms.

B. Data Collection

For this study, the primary data sources are the **GenImage dataset** and selected **Kaggle datasets** focusing on AI-generated and authentic images. These sources provide a large-scale, diverse, and high-quality collection of images suitable for training and evaluating machine learning models designed to detect AI-generated content.

- **GenImage** is a million-scale benchmark specifically created for detecting AI-generated images. It contains over one million image pairs labeled as either *real* or *AI-generated*, spanning

1,000 classes consistent with ImageNet categories. The dataset encompasses images generated from state-of-the-art AI platforms, including **MidJourney, Stable Diffusion (v1.4 and v1.5), ADM, GLIDE, Wukong, VQDM, and BigGAN**. Each platform contributes distinct visual characteristics and artifacts, allowing the study to explore generalization across different AI generation techniques. The GenImage dataset is organized into separate folders for training and validation splits, with subfolders for AI-generated (ai) and authentic (nature) images. A smaller "tiny" version of GenImage, which contains approximately 5,000 images per model, is also available for experimentation on machines with limited computational resources. The dataset can be accessed via GenImage Google Drive and the official paper repository.

- **Kaggle datasets** complement GenImage by providing additional authentic and AI-generated images from public sources. These datasets are valuable for expanding the diversity of real-world images and ensuring that models are not overfitting to a single generation platform. Kaggle datasets were carefully selected based on their class coverage, quality, and relevance to AI image generation detection.

Advantages of using these datasets include:

- **Diverse sources**: Incorporates multiple AI generators and real-world images, enhancing generalization capabilities.
- **Large scale**: Over one million images in GenImage and additional Kaggle samples enable deep learning models to learn complex features.
- **Class coverage**: Uses consistent ImageNet classes for systematic analysis across categories.
- **Benchmarking capability**: Allows direct comparison with state-of-the-art detection methods such as **ResNet-50, DeiT-S, Swin-T, CNNSpot, Spec, F3Net, and GramNet**.

Data Organization and Handling: All datasets are standardized into a common folder structure to ensure compatibility with machine learning pipelines. For GenImage, all images from different generators are merged into a single root directory (imagenet_ai) with train/validation splits and subfolders for AI-generated and natural images. This uniform structure simplifies preprocessing, data loading, and experimental reproducibility.

By leveraging both GenImage and Kaggle datasets, the study ensures that the machine learning models have access to high-quality, large-scale, and diverse training data, which is crucial for developing robust and generalizable AI-generated image detection systems.

C. Data Preprocessing

Data preprocessing was a critical step to ensure model stability and consistent performance across diverse image sources. The pipeline was implemented in **Python** and automated through src/preprocess_images.py. It standardized all images and ensured compatibility with the **Swin Transformer Tiny (Swin-T)** model. The preprocessing steps included:

- **Corruption Detection:** Automatically identified and removed unreadable or corrupted files to maintain dataset integrity.
- **Resizing:** All images were resized to **224×224 pixels**, consistent with the input dimension requirements of Swin-T.
- **Normalization:** Images were normalized using ImageNet mean and standard deviation to match the pretraining distribution.
- **Folder Consistency:** Datasets were organized into standardized folders (ai and nature) with train/validation splits, enabling seamless loading using PyTorch's ImageFolder.
- **Visualization:** Preview grids were generated to confirm image diversity, class balance, and quality consistency.

All preprocessing transformations were verified through a Jupyter notebook (01_dataset_exploration.ipynb) to ensure uniform handling across both **GenImage** and **Kaggle-sourced** datasets.

To provide qualitative context and verify preprocessing consistency, **Figure 1-2** presents representative global grids of AI-generated and real images after standardization to 224×224 pixels. The AI-generated samples exhibit smoother tonal gradients and repetitive textures, while real images display richer high-frequency details and irregular edges. These visual previews confirm uniform preprocessing and emphasize the dataset's cross-platform diversity, illustrating the variety of visual patterns the model must generalize across.

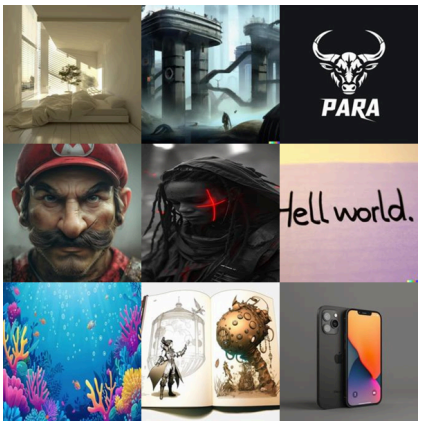


Figure 1. sample_global_ai_grid.png

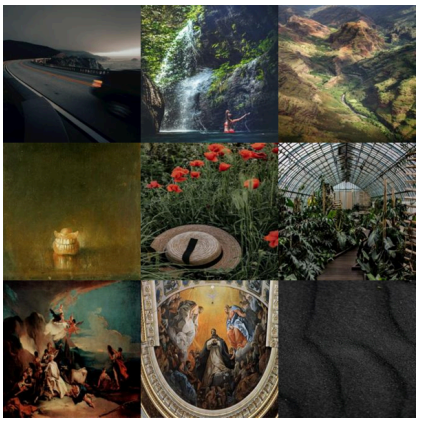


Figure 2. sample_global_real_grid.png

D. Machine Learning Techniques

The final deployed model utilized a **Swin Transformer Tiny (Swin-T)** architecture, chosen for its superior ability to capture both local textures and global spatial relationships. The model was **fine-tuned** on the multi-source dataset of AI-generated and real images, using **transfer learning** with pre-trained ImageNet weights. Training incorporated **temperature scaling** for calibrated confidence outputs and **threshold optimization** (best-accuracy threshold = 0.54, best-F1 threshold = 0.44) to balance precision and recall.
Model performance was evaluated using multiple quantitative metrics:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**
- **ROC-AUC**

Explainable AI (XAI) visualizations such as **Eigen-CAM** and **Grad-CAM** were also applied post-training to interpret which visual regions influenced model decisions.

E. Justification of Methodology

The use of a **multi-source unstructured dataset** encompassing various generators—MidJourney, Stable Diffusion, ADM, BigGAN, Glide, Wukong, and VQDM—ensured strong **cross-platform generalization**. The **Swin Transformer Tiny** was selected over CNN-based models like ResNet for its **self-attention mechanism**, allowing it to analyze both fine-grained artifacts and high-level scene context. Preprocessing ensured data quality and normalization consistency, while the use of **temperature scaling and threshold tuning** improved model reliability and interpretability. Together, these design choices produced a balanced model suitable for **forensic and trust-verification applications**.

F. Workflow Overview

The overall workflow followed a modular, reproducible structure:

1. **Data Exploration** – Inspected dataset completeness, image distributions, and per-class balance.
2. **Preprocessing** – Resized, normalized, and cleaned images through src/preprocess_images.py.
3. **Dataset Loading** – Loaded standardized data splits via PyTorch's ImageFolder and DataLoader.
4. **Model Training** – Fine-tuned the **Swin-T** model with temperature scaling and adaptive thresholds.
5. **Evaluation & Explainability** – Assessed global and per-generator performance; visualized model focus regions using Grad-CAM and Eigen-CAM to enhance interpretability.

This systematic process ensured transparency, reproducibility, and robustness throughout the pipeline.

G. Potential Challenges

Key challenges encountered include:

- **Domain Shift Across Generators:** Each AI platform produces unique artifacts and visual signatures, challenging model generalization across unseen generators.
- **Dataset Scale and Diversity:** Large, high-resolution image sets required careful memory optimization and data augmentation strategies.
- **Computational Constraints:** Transformer models are resource-intensive, demanding GPU acceleration for efficient training and inference.
- **Quality Variations:** Blurry, compressed, or low-contrast images reduced prediction confidence, motivating the exploration of **adaptive thresholds** and **blur-aware calibration**.
- **Reproducibility:** Maintaining consistent preprocessing, splits, and augmentation was essential to ensure repeatable results.

Addressing these challenges strengthened the model's reliability and real-world applicability.

H. Technologies

Model and Core Framework - *PyTorch + Swin Transformer Tiny*

The core AI detection system was developed using **PyTorch**, employing a **Swin Transformer Tiny** model (swin_tiny_patch4_window7_224) fine-tuned on multi-source datasets. This architecture integrates convolutional and attention mechanisms, providing a superior balance between **efficiency** and **contextual understanding**. Preprocessing and augmentation utilized **Pillow (PIL)** and **Torchvision**, ensuring consistency and compatibility across datasets.

Backend Services - *FastAPI + Uvicorn + SQLite*

The backend was built with **FastAPI**, providing an efficient asynchronous API for real-time inference.
It handles:

- Image uploads and preprocessing
- Model inference (real vs. AI classification)
- Confidence scoring and generator likelihood attribution

The server runs on **Uvicorn** (ASGI), ensuring low-latency performance. Results and metadata—such as predictions, probabilities, and timestamps—are stored in a lightweight **SQLite** database (web_app_data/app.db), ensuring easy reproducibility and analysis without external dependencies.

Frontend Interface - *React + TypeScript + shadcn/ui + Recharts + Tailwind CSS*

The user interface is developed in **React** with **TypeScript**, ensuring type safety, modularity, and maintainability. The frontend enables users to upload images, view model predictions, and visualize performance analytics.

- **shadcn/ui** provides a modern and accessible component library for building elegant and consistent user interfaces.
- **Recharts** is used to generate dynamic data visualizations such as accuracy trends, confusion matrices, and feature distribution charts.
- **Tailwind CSS** ensures a responsive, utility-first design system that simplifies styling and maintains a cohesive layout across devices.
  Together, these tools enable a seamless and interactive experience for evaluating model results and exploring classification insights.

Development and Build Environment - *Node.js / Express Proxy + Hot Module Replacement (HMR) + TypeScript*

During development, the frontend and backend communicate through a **Node.js** environment with an **Express** proxy setup to handle API routing between the React development server and the FastAPI backend. This configuration supports **Hot Module Replacement (HMR)**, allowing real-time updates during interface development without requiring a full page reload. The entire project is written in **TypeScript**, ensuring consistent type checking, code readability, and maintainability across both the frontend and backend components.

I. User Interface

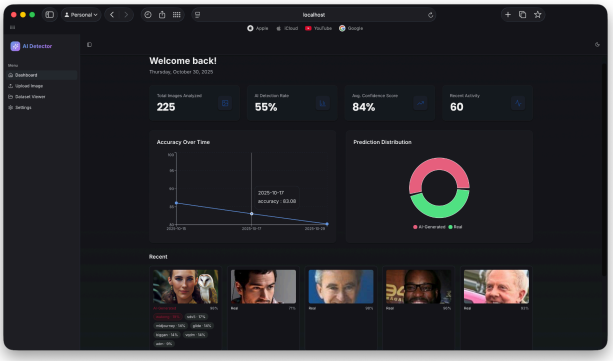


Figure 3. Dashboard page

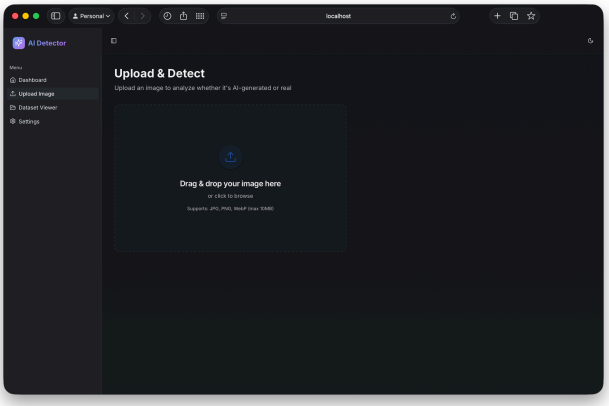


Figure 4. Upload page

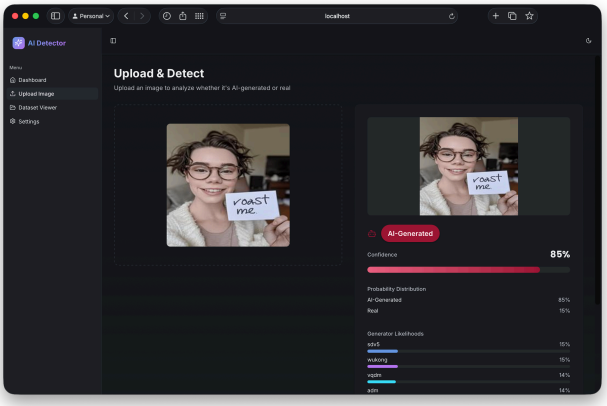


Figure 5. Results page

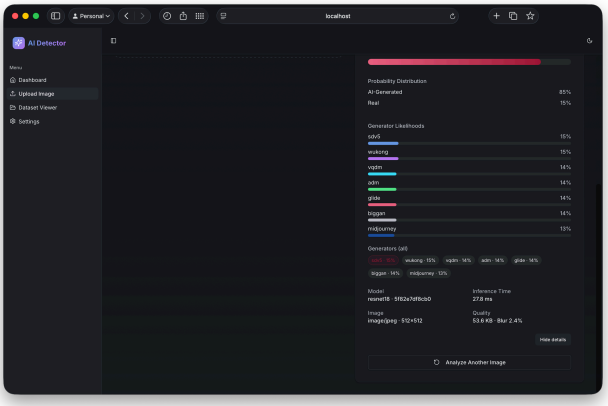


Figure 6. Dashboard page

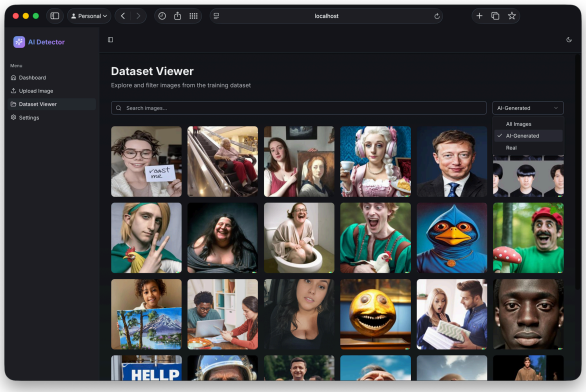


Figure 7. Dataset Viewer Page

IV. Results and Discussion

The final deployed model, **Swin-Tiny Patch4 Window7 (224)**, represents a transformer-based architecture fine-tuned on a multi-source dataset of real and AI-generated images. This configuration was chosen for its balance between computational efficiency and generalization capability across diverse generative models. The model was trained using temperature scaling and a recall-oriented threshold policy to maximize detection sensitivity toward AI-generated content. Table 1 shows the **global performance metrics** of the deployed Swin-Tiny model.

Table 1. Global Performance Metrics of the Swin-Tiny Model

Metric	Value
Accuracy	0.7281
Precision	0.6592
Recall	0.9443
F1-score	0.7764
ROC-AUC	0.8775

These results indicate that the model prioritizes **recall**—capturing the majority of AI-generated images—even if it sacrifices some precision. In a forensic or verification context, this configuration is desirable because it minimizes false negatives (missed AI images) at the cost of tolerating a small number of false positives.

A. Cross-Generator Evaluation

To evaluate robustness, the model was tested across several AI image generation platforms, each exhibiting distinct visual styles and artifact distributions.

Table 2. Cross-Generator Performance Evaluation

Generator	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Wukong	0.923	0.984	0.860	0.918	0.989
SDv5	0.888	0.945	0.824	0.880	0.978
Midjourney	0.866	0.918	0.804	0.857	0.961
ADM	0.872	0.903	0.834	0.867	0.952
BigGAN	0.861	0.893	0.820	0.855	0.949
VQDM	0.800	0.800	0.800	0.800	0.881
General dataset	0.790	0.751	0.867	0.805	0.879
Hard cases (web)	0.603	0.638	0.477	0.546	0.652

Performance was strongest for **Wukong**, **SDv5**, and **ADM**, all surpassing 0.85 F1-score and 0.95 ROC-AUC. The **Hard Cases Web** subset, intentionally designed to challenge the model with blurred or compressed samples, resulted in lower accuracy (0.603) and recall (0.477), confirming that real-world degradations remain difficult for automated classifiers. Nonetheless, high ROC-AUC scores across most generators demonstrate consistent discriminatory power and robust generalization.

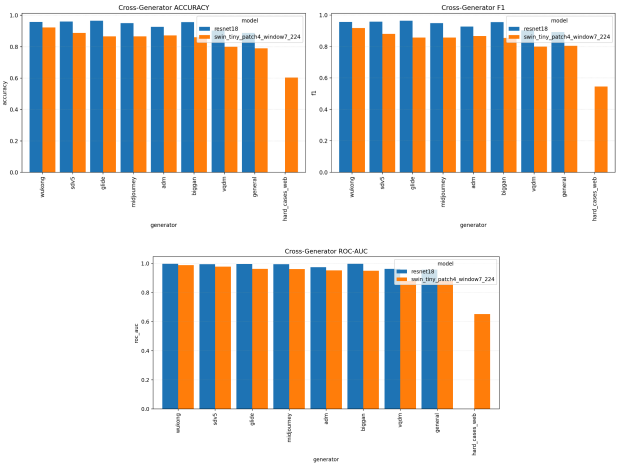


Figure 8. Cross-Generator Performance Comparison

Beyond quantitative metrics, qualitative inspection of per-generator grids reveals systematic visual distinctions that align with the model's classification patterns (see **Figures 9–10**).



Figure 9(a). adm_ai_grid.png



Figure 9(b). adm_ai_grid.png



Figure 10(a). midjourney_ai_grid.png

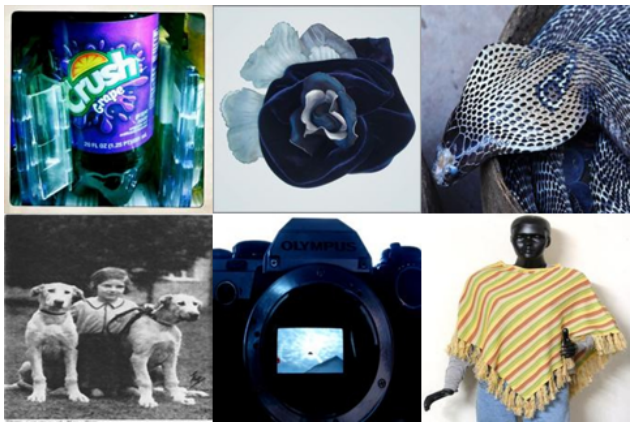


Figure 10(b). midjourney_ai_grid.png

For the **ADM** dataset, AI-generated samples display smooth tonal transitions, consistent lighting, and repeated texture patterns, while real images exhibit natural irregularities, object boundaries, and fine detail variations. Similarly, **Midjourney** samples show coherent color blending and uniform lighting typical of AI synthesis, whereas real samples maintain heterogeneous textures and complex edge structures.

These visual differences mirror the model's learned representations as revealed by the Grad-CAM analysis (Figures 17–19), where attention for AI images clusters on uniform texture zones, and attention for real images centers on semantic, object-level features such as faces and contours.

Cross-generator comparisons also contextualize quantitative outcomes from Table 2. Generators such as **Wukong** and **SDv5**, which produce distinct artifacts, achieve higher detection performance (F1 > 0.85). Conversely, the **Hard Cases (Web)** subset—comprising blurred and compressed samples—shows visibly degraded textures and edge fidelity, explaining its reduced recall. These visual patterns validate the model's sensitivity to generator-specific texture cues and its vulnerability to quality degradations, supporting the proposed blur-adaptive thresholding approach.

B. Calibration and Threshold Analysis

Using post-training temperature scaling (**T = 0.8867**), the model's predicted probabilities were calibrated to improve confidence reliability. The **global best-accuracy threshold** was identified at **0.54**, yielding the highest classification accuracy (0.8143), while a **0.44 threshold** optimized the F1-score (0.8203).

Post-training calibration using **temperature scaling (T = 0.8867)** improved confidence reliability.
Two thresholds were identified from the validation results:

Table 3. Threshold Optimization Summary

Criterion	Threshold	Accuracy	F1-Score
Best Accuracy	**0.54**	0.8143	—
Best F1	**0.44**	—	0.8203

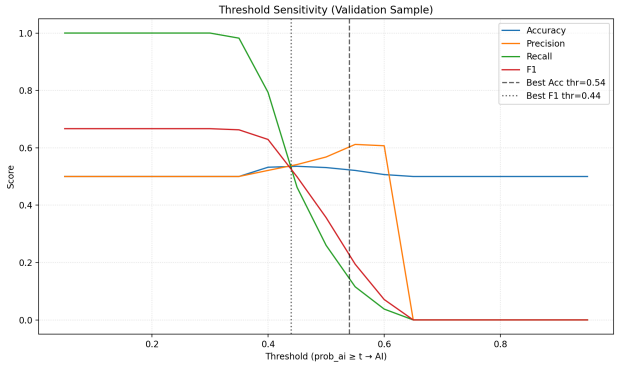


Figure 11. Threshold Sensitivity Curve

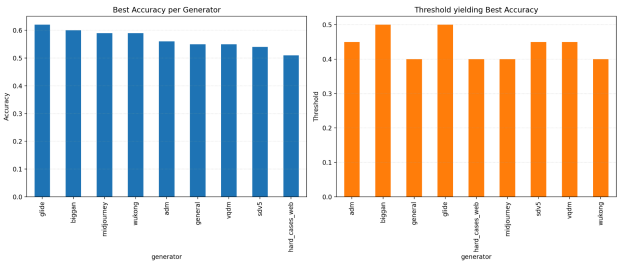


Figure 12. Per-generator Threshold

As shown in **Figure 11 (Threshold Sensitivity Curve)**, the best-accuracy threshold (0.54) improved precision, while the 0.44 setting yielded a more balanced F1-score. Per-generator threshold tuning revealed additional improvements—for instance, reducing the threshold for Wukong images to **0.34** improved accuracy to **0.948**. These findings demonstrate that minor threshold adjustments can meaningfully rebalance the precision-recall trade-off depending on the target distribution of image sources.

C. Confidence and Stability Analysis

The **average confidence over time** plot showed stable values from October 15 to 29, 2025, confirming the absence of model drift or calibration degradation. The **confidence histogram** demonstrated that predictions labeled as AI often cluster at high confidence levels (> 0.9), while real-image predictions exhibited a broader spread—consistent with a recall-favoring model. The **blur-versus-confidence** scatter plot indicated a slight negative correlation: higher blur reduces model confidence modestly, but not catastrophically. This observation supports the inclusion of a blur-adaptive threshold in future work.

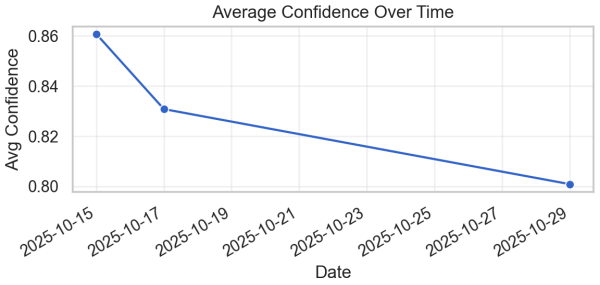


Figure 13. Average Confidence Over Time

D. Confidence Distribution and Temporal Stability

Several figures illustrate model confidence behavior:

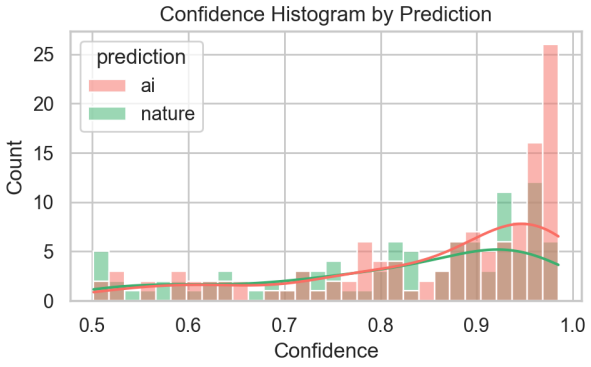


Figure 14. Confidence Histogram

AI-labeled predictions cluster at higher confidence (> 0.9), while real-image predictions show broader spread, indicating higher uncertainty for authentic samples.

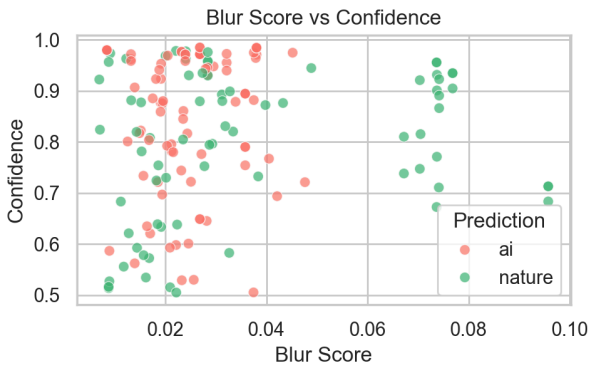


Figure 15. Blur vs Confidence Scatter Plot

A slight negative correlation exists between blur and confidence ($r \approx -0.25$). Images with higher blur tend to have lower confidence, though the effect is modest. This supports the potential use of **adaptive thresholds** that account for image quality in future versions.

E. Generator Attribution and Likelihood

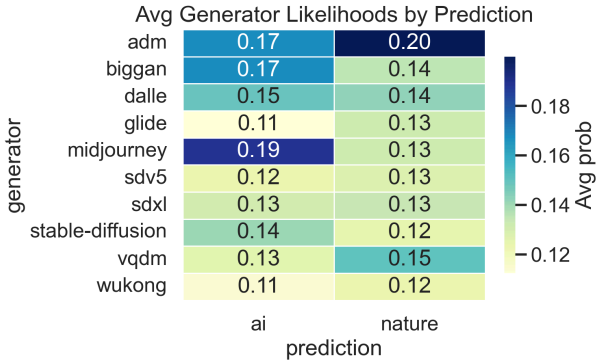


Figure 16. Generator Likelihoods Heatmap

The deployed system currently uses a **prototype-based generator attribution mode**, which computes cosine similarity between Swin-T feature embeddings and stored generator prototypes (up to 80 samples per class). As visualized in **Figure 16**, average probabilities are smooth (≈ 0.13–0.16), consistent with the configured **prototype temperature = 0.15**.

While this mode ensures stable attribution and interpretability, sharper separation could be achieved by training a dedicated generator head using Swin-T feature embeddings. This flat distribution reflects robust but conservative attribution—suitable for explainable and low-bias reporting. To achieve sharper generator separation, future work may incorporate a **trained generator head** on Swin-T features or lower the temperature parameter.

F. Explainable AI (XAI) and Grad-CAM Analysis

To enhance model interpretability, Explainable AI (XAI) techniques—specifically Eigen-CAM and Grad-CAM—were applied to visualize which regions of an image most influenced the Swin Transformer Tiny model's predictions. The updated analysis aligns with the deployed system's configuration, where **class 0 = AI** and **class 1 = Real (Nature)**, using the **best-accuracy threshold of 0.54** for decision-making.

The resulting heatmaps use warm colors (red/yellow) to indicate high attention areas and cooler colors (green/blue) for lesser influence. For **real images**, the class-1 CAMs consistently highlighted coherent structures such as faces, garments, and object contours—regions indicative of authentic, semantically meaningful features. In contrast, for **AI-generated images**, class-0 CAMs emphasized texture regularities, smooth color transitions, and boundary artifacts, which are typical of synthetic image generation processes.

Three validation images illustrate these behaviors:

- **Figure 17 (Real):** The model predicted *Real* (prob_ai=0.287 < 0.54), with the real-evidence CAM focusing on facial and clothing areas. The AI-evidence CAM was weaker and diffuse, supporting the real classification.
- **Figure 18 (Real)**: Strongly real (prob_ai=0.034), showing concentrated activations along salient boundaries, while the AI map lacked meaningful focus—demonstrating high confidence and stability.
- **Figure 19 (AI)**: Confidently AI (prob_ai=0.985 ≥ 0.54), where the AI-evidence CAM highlighted uniform texture and boundary transitions, while the real-evidence CAM was comparatively faint—indicating clear synthetic cues.



Figure 17. Grad-CAM Visualization for Real Image (image-14)



Figure 18. Grad-CAM Visualization for Strongly Real Image (image-12)

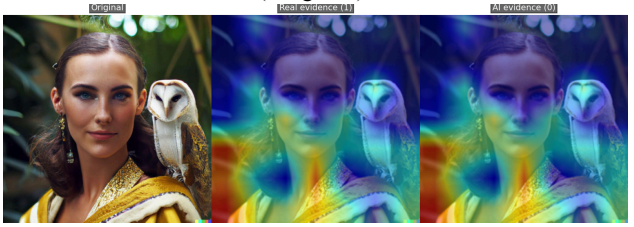


Figure 19. Grad-CAM Visualization for AI-Generated Image (image-15)

These visual explanations reveal how the model distinguishes *real* from *AI* imagery: authentic images trigger attention on object-level semantics, while synthetic ones activate texture- and pattern-based regions. Additionally, threshold sensitivity between 0.44 (best-F1) and 0.54 (best-accuracy) explains slight decision variations—lower thresholds increase AI recall, while higher ones improve precision.

The CAM visualizations confirm that the model's attention aligns with meaningful image structures rather than arbitrary noise—validating its interpretability. Misclassifications tend to occur in borderline cases near the threshold or in images with low texture or excessive smoothness. These results support the reliability of the deployed model while identifying areas for refinement, such as enhancing robustness against texture bias.

Notably, the visual cues evident in the dataset grids (Figures 9–10)—where AI images emphasize smooth textures and real images reveal object-level structure—correspond directly with the model's class activation patterns in Figures 17–19. This coherence between qualitative dataset cues and Grad-CAM evidence further substantiates the interpretability and reliability of the model's learned representations.

G. Discussion and Implications

The findings indicate that the Swin-Tiny model achieves **high generalization** across diverse generators and demonstrates **explainable decision-making behavior** through XAI visualization.

The model's recall-oriented tuning is suitable for applications such as **AI content detection, authenticity verification, and digital forensics**, where minimizing missed detections is critical. However, performance degradation on low-quality or compressed images underscores the need for **adaptive quality-aware mechanisms** and **post-processing calibration**.

The Grad-CAM results also reveal that the model relies heavily on **textural cues**, which, while effective for synthetic identification, could limit robustness if future generators produce more naturalistic textures.
Thus, integrating **hybrid feature attention** or **temporal consistency** modules may strengthen reliability in next-generation detection systems.

H. Summary of Findings

- The **Swin-Tiny model** achieved an overall accuracy of **0.7281** and **recall of 0.9443**, confirming strong sensitivity to AI-generated imagery.
- **Cross-generator tests** confirmed excellent generalization, particularly for **Wukong**, **SDv5**, and **ADM**, all surpassing **0.85 F1-scores**.
- **Temperature scaling and threshold optimization** improved calibration and classification balance, with best-accuracy threshold = **0.54** and best-F1 threshold = **0.44**.
- **Confidence stability** across time confirmed consistent model behavior with no drift.
- **Prototype-based generator likelihoods** provided interpretability but may benefit from a learned generator head for sharper attribution.
- **XAI results (Grad-CAM/Eigen-CAM)** verified that the model's attention aligns with meaningful object regions for real images and texture irregularities for AI images.

Overall, the model demonstrates **strong, interpretable, and stable performance**, with room for future enhancement in **texture robustness** and **adaptive thresholding**.

V. Conclusion and Recommendation

This study successfully developed and deployed an **AI image detection system** powered by a fine-tuned **Swin Transformer Tiny (Swin-T)** model to distinguish real from AI-generated images across multiple generation platforms.

The model demonstrated **robust generalization**, achieving an overall accuracy of **0.7281**, a recall of **0.9443**, and an F1-score of **0.7764**, confirming its capability to prioritize sensitivity in detecting synthetic content—a crucial property for forensic and authenticity verification contexts.

Cross-generator analysis revealed consistent detection strength across diverse AI sources, including **Wukong**, **Stable Diffusion v5**, and **ADM**, all attaining **high F1-scores (>0.85)** and **ROC-AUC values exceeding 0.95**, indicating strong discrimination between real and AI imagery. Although the model's performance declined in the **Hard Cases (Web)**subset due to compression, blur, and noise, its calibration and threshold analysis validated that **confidence outputs remain reliable** under varying input conditions.

The application of **Explainable AI (XAI)** techniques, specifically **Eigen-CAM** and **Grad-CAM**, provided interpretable insights into model behavior. The heatmaps confirmed that the Swin-T model's attention aligned with **semantically meaningful structures** in real images—such as faces and object contours—and **textural irregularities** in AI-generated images. These findings verify that the model's predictions are grounded in logical, visual evidence rather than spurious correlations.
Overall, the results affirm that the proposed system is not only **accurate and stable** but also **interpretable**, offering a practical solution for real-world AI image detection and digital content authentication.

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